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$$\rho^{-1}(\alpha(s) - c) = \rho^{-1}(\rho(\bar{\alpha}(s))) = \bar{\alpha}(s) \quad \cdots\cdots (65)$$

Finally, ρ^{-1} is linear so that

$$\rho^{-1}(\alpha(s) - c) = \rho^{-1}(\alpha(s)) - \rho^{-1}(c) \quad \cdots\cdots (66)$$

Comparing (65) and (66), one obtains,

$$\bar{\alpha}(s) = \rho^{-1}(\alpha(s)) - \rho^{-1}(c) \quad \cdots\cdots (67)$$

(63) reveals the fact that the 2 curves α and $\bar{\alpha}$ are related by a rigid motion

□

- We will prove the existence part of the Fundamental theorem in the next class.

Lecture 3

Choose $s_0 \in I$, a point $p_0 \in \mathbb{R}^3$, and a positively oriented orthonormal initial frame $\{T_0, N_0, B_0\}$ with $F_0 = [T_0, N_0, B_0]$ being an orthogonal matrix of unit determinant. We now have to construct a curve from prescribed $k(s) > 0$ and $\tau(s)$.

Define
$$A(s) = \begin{bmatrix} 0 & -k(s) & 0 \\ k(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{bmatrix} \quad \cdots\cdots (1)$$

Check that $A(s)^T = -A(s)$

Solve the matrix differential equation $F'(s) = F(s)A(s)$ with $F(s_0) = F_0$ $\quad \cdots (2)$

$$\begin{bmatrix} T_1'(s) & N_1'(s) & B_1'(s) \\ T_2'(s) & N_2'(s) & B_2'(s) \\ T_3'(s) & N_3'(s) & B_3'(s) \end{bmatrix} = \begin{bmatrix} T_1(s) & N_1(s) & B_1(s) \\ T_2(s) & N_2(s) & B_2(s) \\ T_3(s) & N_3(s) & B_3(s) \end{bmatrix} \begin{bmatrix} 0 & -k(s) & 0 \\ k(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{bmatrix} \quad \cdots (3)$$

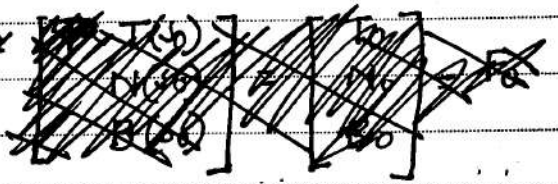
Explicitly

$$\begin{bmatrix} T_1'(s) & N_1'(s) & B_1'(s) \\ T_2'(s) & N_2'(s) & B_2'(s) \\ T_3'(s) & N_3'(s) & B_3'(s) \end{bmatrix} = \begin{bmatrix} k(s)N_1(s) & -k(s)T_1(s) - \tau(s)B_1(s) & \tau(s)N_1(s) \\ k(s)N_2(s) & -k(s)T_2(s) - \tau(s)B_2(s) & \tau(s)N_2(s) \\ k(s)N_3(s) & -k(s)T_3(s) - \tau(s)B_3(s) & \tau(s)N_3(s) \end{bmatrix}$$

By equating the columns, one has

$$\left.\begin{aligned} T'(s) &= k(s)N(s) \\ N'(s) &= -k(s)T(s) - \tau(s)B(s) \\ B'(s) &= \tau(s)N(s) \end{aligned}\right\} \cdots [4)$$

In other words, we are looking for the unique solution of the linear differential system with continuous coefficients with the initial value $F(s_0) = [T(s_0), N(s_0), B(s_0)] = [T_0, N_0, B_0] = F_0$



Now, $\frac{d}{ds}\left(F(s)F(s)^T\right) = F'(s)F(s)^T + F(s)\left(F'(s)\right)^T$

$= F(s)A(s)F(s)^T + F(s)A^T(s)F(s)^T$; [By (2)]

$= F(s)\left(A(s) + A^T(s)\right)F(s)^T$

$= 0$; $[\because A^T(s) = -A(s)]$

Thus $F(s)F(s)^T = F_0(s)F_0(s)^T = I_3$ ..... (5)

$\therefore F^{-1}(s) = F(s)^T$. Hence $F(s)F(s)^T = F(s)^T F(s) = I_3$. Since $F(s)$ is an orthogonal matrix, its columns are orthonormal.

Now the function $g(s) = \det F(s)$ ..... (6)

is continuous because the entries of $F(s)$ are continuous and the determinant is a polynomial in those entries.

We know: $g(s) \in \{-1, 1\}$ for every $s \in I$, $g(s_0) = 1$

If $g(s_1) = -1$ anywhere in I, the intermediate value theorem on the interval between s_0 and s_1 would give

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some s_* with $g(s_*) = 0$,

a contradiction.

Hence, $\boxed{\det F(s) = 1 \quad \forall s \in I}$ ..... (7)

- Note that $s_0 \in I$ is just a prescribed parameter value at which the initial data $F(s_0)$ is given. It doesn't mean that the parameter s is supposed to be always greater or equal to s_0.

Now define $\alpha(s) = p_0 + \int_{s_0}^{s} T(u)\, du$ ..... (8)

Then by the fundamental theorem of calculus,

$$\alpha'(s) = T(s) \quad \text{..... (9)}$$

And since $\{T(s), N(s), B(s)\}$ constitutes an orthonormal basis at each parameter value $s \in I$,

$$|\alpha'(s)| = |T(s)| = 1 \quad \text{..... (10)}$$

Thus the curve $\alpha(s)$ defined by (8) is arc-length parametrized and hence regular.

Furthermore, $\alpha''(s) = T'(s)$

$$= k(s) N(s); \ [\text{By (4)}] \quad \text{..... (11)}$$

Since $|N(s)| = 1$ by orthonormality of the basis $\{T(s), N(s), B(s)\}$ and $k(s) > 0$, $\forall s \in I$, one obtains

$$|\alpha''(s)| = k(s), \quad \frac{\alpha''(s)}{|\alpha''(s)|} = N(s) \quad \text{..... (12)}$$

Hence, the curvature of $\alpha(s)$ is indeed prescribed by $k(s)$ and its principal normal is the constructed vector $N(s)$. Now, since $\{T(s), N(s), B(s)\}$ constitutes an orthonormal system and $\det F(s) = 1$, one must have

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$$\vec{B}(s) = \vec{T}(s) \wedge \vec{N}(s) \quad \cdots\cdots\cdots\cdots (13)$$

So, the binormal of the curve $\alpha(s)$ is the constructed vector $B(s)$.

Finally, by (4), we have $B'(s) = \tau(s) N(s)$,

so the torsion of $\alpha(s)$ is precisely the prescribed function $\tau(s)$.

This proves existence and completes the fundamental theorem.

QED

"End of lecture 3"

Curve	Base point at $s=0$	Frenet tripod
original curve α	$p_0 = \alpha(0)$	$\{T_0, N_0, B_0\}$
Normalized curve $\beta = M \circ \alpha$	$0 = \beta(0)$	$\{e_1, e_2, e_3\}$

Lecture 4

The local canonical form

The key idea is to fix the Frenet tripod at one point and use it as a coordinate system for the nearby curve.

Let $\alpha: I \to \mathbb{R}^3$ be a curve parametrized by arc-length without singular points of order 1. k is a differentiable function with $k(s_0) > 0$. By continuity of k, we have $k > 0$ in a nbhd of s_0, so that the Frenet frame is defined there. Put $h = s - s_0$ and write $\tilde{\alpha}(h) = \alpha(s_0 + h) \quad \cdots\cdots (1)$

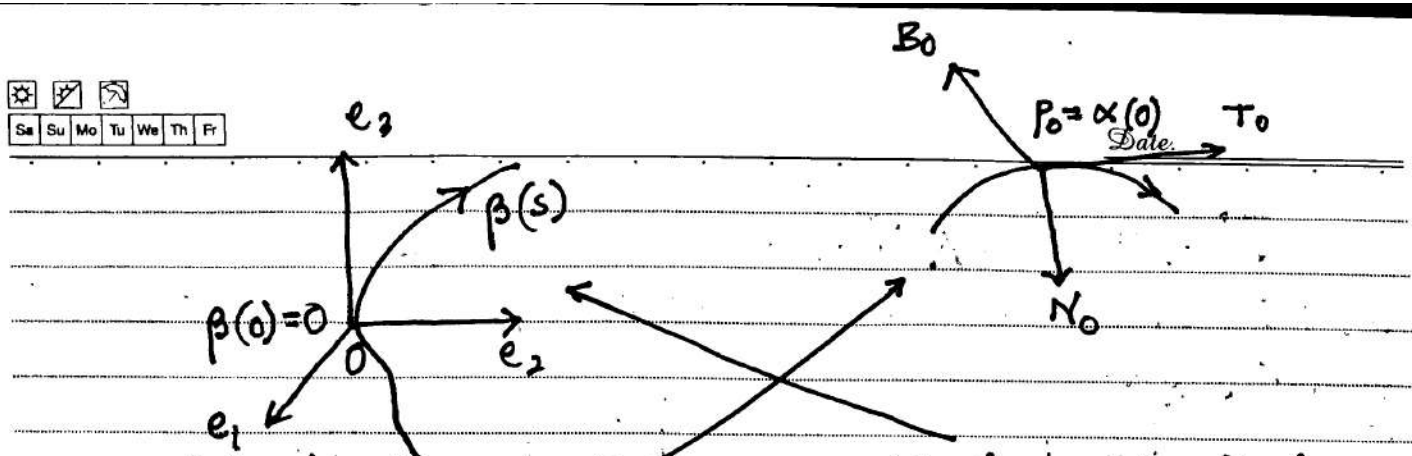
Then $\tilde{\alpha}'(h) = \alpha'(s_0+h) \frac{d}{dh}(s_0+h) = \alpha'(s_0+h) \quad \cdots\cdots (2)$

Thus $|\tilde{\alpha}'(h)| = 1$; [$\because |\alpha'(s)| = 1$ as α is arc-parametrized]

Write $p_0 = \alpha(0)$ (we dropped the tilde and we'll be denoting s by h from now on)

$T_0 = T(0)$, $N_0 = N(0)$, $B_0 = B(0) \quad \cdots\cdots\cdots (3)$

We place the spatial origin at p_0 and use T_0, N_0, B_0 as the positive x, y, z directions, resp.



- source tripod : $\{T_0, N_0, B_0\}$
  - located at $P_0 = \alpha(0)$
  - $\beta(s) = M(\alpha(s))$ passes through the origin
- Destination tripod : $\{e_1, e_2, e_3\}$
  - located at the origin O
  - $\alpha(s)$ passes through $\alpha(0) = P_0$

In the uniqueness proof: the source tripod $\{\bar{T}_0, \bar{N}_0, \bar{B}_0\}$ located at $\bar{P}_0 = \bar{\alpha}(s_0)$ was transported to the destination tripod $\{T_0, N_0, B_0\}$ located at $P = \alpha(s)$ with the rotation ρ represented by the matrix R with $R = F_0 \bar{F}_0^T$, where $F_0 = [T_0, N_0, B_0]$ and $\bar{F}_0 = [\bar{T}_0, \bar{N}_0, \bar{B}_0]$ being 3×3 matrices that are both orthogonal with unit determinant. Here, $\rho(\bar{T}_0) = T_0$, $\rho(\bar{N}_0) = N_0$, $\rho(\bar{B}_0) = B_0$. with $M(\bar{\alpha}(s_0)) = \alpha(s)$

In the present case, $M(P_0) = O$, $\rho(T_0) = e_1$, $\rho(N_0) = e_2$, and $\rho(B_0) = e_3$ ------(4)

And the rotation matrix $R = F_0^T$ simply with

$F_0 = [T_0, N_0, B_0]$ ------(5)

For our present purpose, we denote the matrix F_0 by S so that $R = S^T$ with $S = [T_0, N_0, B_0]$ ---------- (5)

We'll call $\beta(s)$ with $\beta(0) = O$, the <u>normalized curve</u> defined by $\beta(s) = M(\alpha(s)) = \begin{bmatrix} x(s) \\ y(s) \\ z(s) \end{bmatrix}$ ---------- (6)

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We first rotate about $p_0 = \alpha(0)$ and then translate it to the origin.

$$R_{p_0}(x) = p_0 + \rho(x - p_0) \quad \ldots\ldots\ldots (7)$$

$\therefore \xi_1(s) = p_0 + \rho(\alpha(s) - p_0)$ (rotate about p_0)

$\beta(s) = \xi_1(s) - p_0$ (translate by $-p_0$)

$= p_0 + \rho(\alpha(s) - p_0) - p_0 = \rho(\alpha(s) - p_0) \ldots\ldots (8)$

Now, $x(s) = \langle \beta(s), e_1 \rangle$

$= \langle \rho(\alpha(s) - p_0), \rho(T_0) \rangle$: [using (4)]

$= \langle \alpha(s) - p_0, T_0 \rangle$: [Because ρ is orthogonal]

Similarly, $y(s) = \langle \alpha(s) - p_0, N_0 \rangle$, $z(s) = \langle \alpha(s) - p_0, B_0 \rangle$

Together
$$\begin{bmatrix} x(s) \\ y(s) \\ z(s) \end{bmatrix} = \begin{bmatrix} \langle \alpha(s) - p_0, T_0 \rangle \\ \langle \alpha(s) - p_0, N_0 \rangle \\ \langle \alpha(s) - p_0, B_0 \rangle \end{bmatrix} \quad \ldots\ldots\ldots (9)$$

We have, $\alpha'(s) = T(s)$, $\alpha''(s) = T'(s) = k(s)N(s) \ldots\ldots (10)$

$N'(s) = -\tau(s)B(s) - k(s)T(s) \ldots\ldots (11)$

Now, differentiate $\alpha''(s) = k(s)N(s)$ given by (10) to obtain

$$\alpha'''(s) = \frac{d}{ds}(k(s)N(s)) = k'(s)N(s) + k(s)N'(s)$$

$$= k'(s)N(s) + k(s)[-\tau(s)B(s) - k(s)T(s)]$$

$$= -[k(s)]^2 T(s) + k'(s)N(s) - k(s)\tau(s)B(s)$$

$\ldots\ldots\ldots (11)$

Set $k_0 = k(0)$, $k_0' = k'(0)$, $\tau_0 = \tau(0) \ldots\ldots (12)$

Evaluating after differentiating gives,

$$\boxed{\begin{aligned} &\alpha'(0) = T_0, \\ &\alpha''(0) = k_0 N_0, \\ &\alpha'''(0) = -k_0^2 T_0 + k_0' N_0 - k_0 \tau_0 B_0 \end{aligned}} \quad \ldots\ldots (13)$$

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- **Taylor expansion**

$$\alpha(s) = \alpha(0) + s\alpha'(0) + \frac{s^2}{2}\alpha''(0) + \frac{s^3}{6}\alpha'''(0) + r_3(s)$$

Taylor remainder

with $\dfrac{|r_3(s)|}{|s|^3} \to 0$

Remainder calculation

$$\frac{1}{2}\int_0^s (s-u)^2 \alpha'''(u)\,du = \left[\frac{(s-u)^2}{2}\alpha''(u)\right]\Big|_0^s + \int_0^s (s-u)\alpha''(u)\,du$$

; [Integration by parts]

$$= -\frac{s^2}{2}\alpha''(0) + [(s-u)\alpha'(u)]\Big|_0^s + \int_0^s \alpha'(u)\,du$$

$$= -\frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - \alpha(0) = -\frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - r_0$$

..... (14)

[illegible]

Now, $r_3(s) = \dfrac{1}{6}\int_0^s (s-u)^3 \alpha^{(4)}(u)\,du$ ..... (15)

Now, $\dfrac{d}{du}\left[\dfrac{(s-u)^3}{6}\right] = -\dfrac{(s-u)^2}{2}$, ... (16)

i.e., $\dfrac{1}{6}d(s-u)^3 = \dfrac{(s-u)^2}{2}du$

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$$\therefore r_3(s) = \left[\frac{(s-u)^3}{6}\alpha'''(u)\right]\Big|_0^s + \frac{1}{2}\int_0^s (s-u)^2\alpha'''(u)\,du$$

$$= -\frac{s^3}{6}\alpha'''(0) - \frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - P_0$$

; [By using (4)]

since $\alpha(0) = P_0$ ..... (17)

$$\therefore \alpha(s) = P_0 + s\alpha'(0) + \frac{s^2}{2}\alpha''(0) + \frac{s^3}{6}\alpha'''(0) + r_3(s) \quad \text{..... (18)}$$

with $r_3(s) = -\frac{s^3}{6}\alpha'''(0) - \frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - P_0$

Remainder estimate:

$$r_3(s) = \frac{1}{6}\int_0^s (s-u)^3\alpha^{(4)}(u)\,du$$

Substitute $u = sv$

u	v
0	0
s	1

Then $r_3(s) = \frac{s^4}{6}\int_0^1 (1-v)^3\alpha^{(4)}(sv)\,dv$

Since α is smooth, take C_4 to be a bound for $|\alpha^{(4)}|$ on a small closed interval about zero.

Now, $\|r_3(s)\| = \frac{|s|^4}{6}\left\|\int_0^1 (1-v)^3\alpha^{(4)}(sv)\,dv\right\|$

$$\leq \frac{|s|^4}{6}\int_0^1 \|(1-v)^3\alpha^{(4)}(sv)\|\,dv$$

$$= \frac{|s|^4}{6}\int_0^1 (1-v)^3\|\alpha^{(4)}(sv)\|\,dv$$

$$\leq \frac{C_4|s|^4}{6}\int_0^1 (1-v)^3\,dv = \frac{C_4|s|^4}{24} \quad \text{..... (19)}$$

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In particular, $\frac{\|r_3(s)\|}{|s|^3} \leq \frac{C_4|s|}{24} \to 0$ ..... (20)

• Now (18) reads off as

$$\alpha(s) - p_0 = s\alpha'(0) + \frac{s^2}{2}\alpha''(0) + \frac{s^3}{6}\alpha'''(0) + r_3(s)$$

$$= sT_0 + \frac{s^2}{2}k_0N_0 + \frac{s^3}{6}(-k_0^2T_0 + k_0'N_0 - k_0\tau_0B_0) + r_3(s)$$

; [By applying (13)]

$$= \left(s - \frac{k_0^2 s^3}{6}\right)T_0 + \left(\frac{k_0}{2}s^2 + \frac{k_0'}{6}s^3\right)N_0 - \frac{k_0\tau_0}{6}s^3B_0 + r_3(s)$$ ..... (21)

Apply ρ on (21) and then applying (8)

$$\beta(s) = \rho(\alpha(s) - p_0) = \left(s - \frac{k_0^2 s^3}{6}\right)\rho(T_0) + \left(\frac{k_0}{2}s^2 + \frac{k_0' s^3}{6}\right)\rho(N_0)$$

$$- \frac{k_0\tau_0}{6}s^3\rho(B_0) + \rho(r_3(s))$$

↑ Normalized curve

$$= \left(s - \frac{k_0^2 s^3}{6}\right)e_1 + \left(\frac{k_0}{2}s^2 + \frac{k_0' s^3}{6}\right)e_2$$

$$- \frac{k_0\tau_0}{6}s^3 e_3 + \rho(r_3(s))$$ ..... (22)

Define r_x, r_y, r_z to be the components (numbers) of the rotated remainder $\rho r_3(s) = r_x(s)e_1 + r_y(s)e_2 + r_z(s)e_3$ .... (23)

$$r_x(s) = \langle \rho r_3(s), e_1 \rangle = \langle \rho(r_3(s)), \rho T_0 \rangle$$

$$= \langle r_3(s), T_0 \rangle$$

But $|r_x(s)| = |\langle r_3(s), T_0\rangle| \leq \|\rho(r_3(s))\| = \|r_3(s)\|$

Similarly $|r_y(s)| \leq \|r_3(s)\|$

$|r_z(s)| \leq \|r_3(s)\|$ } - (24)

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Hence, we can rewrite (22) as

$$\beta(s) = \left(s - \frac{k_0^2 s^3}{6} + r_x(s)\right) e_1 + \left(\frac{k_0}{2} s^2 + \frac{k_0'}{6} s^3 + r_y(s)\right) e_2 + \left(-\frac{k_0 \tau_0}{6} s^3 + r_z(s)\right) e_3$$

"canonical form"

From (6), one obtains

$$\boxed{\begin{aligned} x(s) &= s - \frac{k_0^2}{6} s^3 + r_x(s) \\ y(s) &= \frac{k_0}{2} s^2 + \frac{k_0'}{6} s^3 + r_y(s) \quad , \quad |r_i(s)| \le C_4 \frac{|s|^4}{24} \\ z(s) &= -\frac{k_0 \tau_0}{6} s^3 + r_z(s) \qquad : [\text{By (24) \& (20)}] \end{aligned}} \quad \cdots (25)$$

• Using the same remainders in the plane calculation

Recall that tb plane is called the rectifying plane and the tn plane is called the osculating plane.

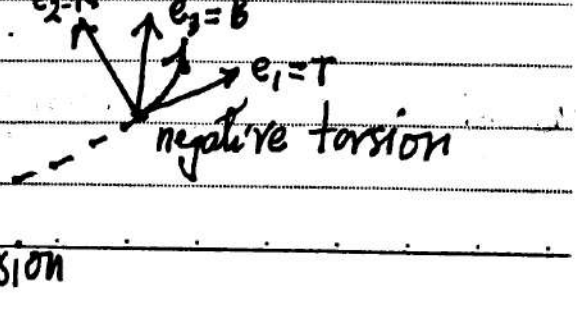
In the normalized coordinates (the frame being $\{e_1, e_2, e_3\}$) the rectifying plane is the $e_1 e_3$ plane, i.e., $y=0$ plane and the osculating plane is the $e_1 e_2$ plane, i.e., $z=0$ plane.

From (25), one then obtains,

$$\left.\begin{aligned} \frac{y(s)}{s^2} &= \frac{k_0}{2} + \frac{k_0'}{6} s + \frac{r_y(s)}{s^2} \to \frac{k_0}{2} > 0 \\ \frac{z(s)}{s^3} &= -\frac{k_0 \tau_0}{6} + \frac{r_z(s)}{s^3} \to -\frac{k_0 \tau_0}{6} \end{aligned}\right\} \cdots (26)$$

Thus $y(s) > 0$ for all sufficiently small $s \neq 0$, on both sides of zero. If $\tau_0 \neq 0$, $z(s)$ changes sign at $s=0$ because s^3 changes sign. In do Carmo's convention, negative torsion gives a crossing toward the positive binormal side

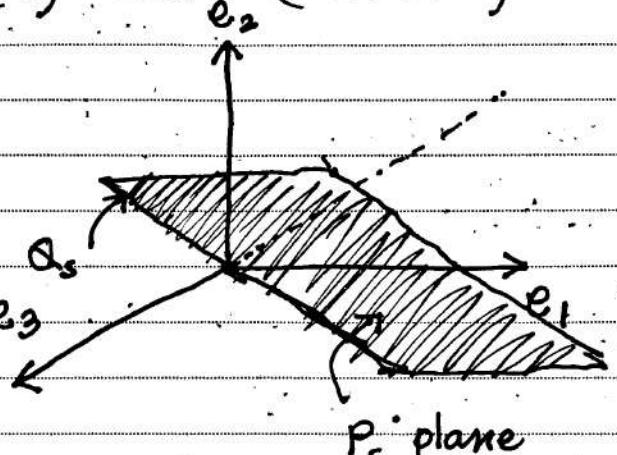
And positive torsion gives a crossing toward the negative binormal side.



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- In our normalized coordinates, the tangent line is the X-axis; i.e., e_1. The plane P_s contains $\beta(s) = (x(s), y(s), z(s))$ and $(x(s), 0, 0)$



The line $Q_s = (0, y(s), z(s))$ lies in the P_s plane. But since its X-coordinate is 0, the line lies in the YZ plane, i.e., The plane P_s intersects the YZ plane at the line Q_s. In fact, the P_s plane is the one that contains the tangent vector e_1 at the origin and a nearby point $\beta(s)$ of the normalized curve.

Now $y(s) > 0$ for small $s \neq 0$. Dividing Q_s by $y(s)$ changes its length, not its direction.

Thus $$P_s = \text{Span}\left\{ e_1, \left(0, 1, \frac{z(s)}{y(s)}\right) \right\} \quad \cdots\cdots (27)$$

So, looking along the x-axis, the cross-section of P_s is the line Q_s with slope $$c(s) = \frac{z(s)}{y(s)} \quad \cdots\cdots (28)$$

The Canonical form (28) gives

$$\frac{y(s)}{s^2} \to \frac{k_0}{2} > 0, \quad \frac{z(s)}{s^3} \to -\frac{k_0 \tau_0}{6}$$

as given by (26). Therefore, $$c(s) = \frac{z(s)}{y(s)} = s \frac{z(s)/s^3}{y(s)/s^2} \quad \cdots\cdots (29)$$

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From (26), one obtains $\frac{z(s)/s^3}{y(s)/s^2} \to -\frac{T_0}{3}$ as $s \to 0$

so that $c(s) \to 0$ as $s \to 0$; [From (29)]

Hence, the second direction $(0, 1, c(s))$ in (27) tends to e_2, while the first direction is always e_1. The limiting plane is therefore the XY plane, $z = 0$, or the osculating plane.

"End of lecture 4"

- Next lecture: global properties of plane curves.